

SELF-INTRODUCTION

This is Edwin, currently working as a Deep Learning Algorithm Engineer leading the company's AI agent infrastructure (daemon / harness / skills / MCP) from the ground up, while serving as CTO for a side venture where I lead technical decision-making and architecture design. My research focused on computer vision and model optimization, with a first-author paper published at ICICML 2023 on knowledge distillation in deep reinforcement learning. Previously, I worked on the Azure Cloud Architecture team at Microsoft and led data analytics and full-stack development at FNZ. I am proficient with PyTorch, TensorFlow, LangGraph, and various AI Agent frameworks including Claude, Codex, DeepSeek, and Multica. I look forward to combining my research background and engineering experience to deliver impactful AI solutions.

EDUCATION

Northeastern University, ME, USA

M.S. in Computer Science

Jan 2024 - Jun 2025

Overall GPA: **4.0**

The Ohio State University, OH, USA

B.S. in Electrical & Computer Engineering

Aug 2015 - Jun 2019

Minor in Mathematics

EXPERIENCE

Shanghai Deep Star Energy Technology Co., Ltd.

Deep Learning Algorithm Engineer

Aug 2026 - Present

Shanghai, China

- Super-agent infrastructure from scratch: built the company's in-house agent stack on top of frontier models (Claude, Codex, DeepSeek) through multi-model iterative loops, owning four core components — daemon (always-on daemon covering task push, session resumption, approval gate and crash recovery), harness (layered rule set and safety floor), skills (reusable skill library with on-demand loading), and MCP (tool and protocol access layer).
- Agent runtime and protocol layer: designed the multi-turn agent loop, tool calling, context compaction & continuation, and layered prompt injection for long-running and cross-session execution; shipped a production-grade MCP service with 8 read-only tools covering service accounts, multi-tenant boundaries, audit and trace correlation, rate limiting, bounded concurrency, fault injection & recovery and a kill switch, validated against real Claude and Codex clients plus a security live matrix.
- AI R&D fleet: orchestrated the toolchain (Paseo, Multica, Metathinking, VS Server, Sol-Pi) into a multi-agent × multi-model × multi-role delivery pipeline spanning architecture review, implementation, verification and release; defined task-as-single-source-of-truth conventions (role routing, evidence chain, reviewer ≠ author) so agent output stays auditable, handover-ready and reviewable.
- Token efficiency & delivery at scale: established four-bucket token metering (input / output / cache read / cache write) and per-task cost ledgers, and continuously improved token utilization via model routing, prompt-cache prefix alignment, layered and on-demand context loading, and deterministic terminal-output compaction; on a single project, delivered 11 pull requests and 9,500 lines of change within 9 calendar days, with test code accounting for 49% of all churn.
- Data platform & AI expert diagnosis: built a data analytics platform for company projects, unifying heterogeneous sources (industrial PLC hourly data, water-supply non-revenue water and DMA zones, project intelligence) and driving an end-to-end natural-language “retrieve-analyze-draft” workflow that delivers AI expert diagnosis reports; representative outputs include pipeline leakage and DMA NRW analysis, online water-meter anomaly detection with meter-replacement planning, and a water project intelligence CRM.
- 2026 “Shuqing Cup” Data Utilization Track (Team Lead / Algorithm Lead): represented the company in the Shanghai Data Utilization and Security Governance Skills Competition, coordinating goal breakdown and milestones for a 5-person team; led the algorithm design of the “Shuimai Zhiqing” water-utility project intelligence product — structured document extraction, cross-source entity resolution, opportunity scoring, gold-standard dataset and model evaluation — delivering a closed loop of cross-platform merging + source verifiability + full-stage tracking + intelligent scoring.

Digital Factory (Side Business)

Co-founder & CTO

Jan 2026 - Present

Shanghai, China

- Lead the construction of a distributed AI factory system based on multi-agent collaboration, responsible for overall technical architecture design and AI model orchestration.
- Design and implement hybrid cloud-on-premises inference architecture, integrating multiple large models including Claude Code, DeepSeek V4 Pro, and Qwen (distilled from Opus 4.7). Achieve local low-cost inference through Ollama, optimizing response speed and computing costs.
- Build a multi-Agent collaboration framework using Multica + OpenClaw, scheduling digital employees like Hermes and QoderWake to process complex tasks in parallel, enabling an automated pipeline for game design operations.
- Establish the technical infrastructure from scratch, covering core modules including AI inference, task orchestration, and multi-model routing.
- Expand the business from game design to multiple verticals including group buying, demonstrating strong system portability and horizontal scalability.

FNZ Information Technology

Analyst Developer

Dec 2021 - July 2022

Shanghai, China

- Worked as a full-stack engineer in agile software development with a focus on the Core Repository - FNZOneX.
- Added a new Advisor group to the existing Administrator and Guest groups & modified Declaration scripts generated by Alchemist from the user permission matrix.
- Embedded the third-party Addressy address lookup API into all address fields on the website.
- Produced runtime SQL script sets for multiple development/test environments.
- Debugged JavaScript errors via browser developer tools (Elements, Console, Sources and Network); utilized Visual Studio debug mode to trace aspx/aspx.vb exceptions.
- Set up, modified and shared IIS server configurations for development and demonstration.
- Collaborated with the global IT department and cross-team peers to resolve cross-region permission and ownership boundary issues.
- Resolved blockers and release issues reported by testers, and formulated the testing procedure prior to functional review.

Wicresoft Co. Ltd. (Microsoft GCR)

Technical Cloud Engineer & Technical Advisor

Oct 2019 - Dec 2021

Shanghai, China

- Provided technical support for four Azure products: Monitor, Log Analytics, Automation & Application Insights.
- Customized alert configuration, query search, data sampling & performance counter collection.
- Captured & analyzed NetMon, ProcMon & Fiddler traces to troubleshoot agent, hybrid worker and portal issues; visualized data through Azure dashboards, Metric Explorer & workbooks.
- Debugged Application Insights SDK configuration scripts in .Net/.Net Core/Java/Python/JavaScript with Visual Studio/VS Code.
- Helped design, modify and debug PowerShell/Python/PS workflow/graphical Runbooks & DSC configurations; advised on partner integration and workspace solutions such as Update Management, Auto Start/Stop VMs, Change Tracking and Inventory.
- Served key clients such as Adobe, Intel, and Amazon and handled high-risk or long-pending case escalations.
- Managed technical coordination for the 12-member team across Shanghai, Wuxi and Beijing; resolved cross-team scoping conflicts.
- Partnered with the team lead and partner technical advisors on case analysis, hiring and team planning.

PROJECT

Industrial AI Data Foundation & Business System AI Transformation Jun 2026 - Jul 2026

Senior AI Development Engineer

Qunxian Group

- Designed and built an industrial-grade AI data foundation, integrating multi-source heterogeneous production data from the Guilin plant (4×42 MVA submerged-arc furnace ferromanganese lines), owning model training and performance optimization.
- Led the intelligent transformation of the ERP system, embedding AI capabilities into supply chain management and resource allocation.
- Drove the AI upgrade of the OA platform, integrating intelligent approval, knowledge Q&A and process automation.

Dataset Distillation of Galaxy CNN

Sep 2024 - Dec 2025

Advisor: Tala Talaei Khoei

NEU

- Demonstrated superior performance in tasks like galaxy morphology classification by using a distilled and curated version of the newest Galaxy Zoo dataset.
- Careful tuning of statistical significance thresholds for validation loss trends to optimize performance.
- Applied Self-Adaptive Trajectory Matching (STM) to prevent overfitting and accelerate the learning process.
- Offered higher classification accuracy and more efficient training, while mitigating class imbalance and improving overall data quality.

Knowledge Distillation of Deep Reinforcement Learning

Jul 2023 - Sep 2023

Advisor: Mark Vogelsberger

MIT

- Built teacher and student DQN models for deep reinforcement learning using PyTorch.
- Carried out comparative experiments in CartPole, MountainCar, Acrobot, LunarLander and Pendulum environments from Gymnasium.
- Systematically tuned hyperparameters (temperature, regression rounds) to evaluate knowledge distillation efficiency across different MDPs.
- Extended knowledge distillation to Deep Recurrent Q-Networks (DRQN) for more complex sequential decision-making tasks.
- The core findings of this project were published as a full paper at ICICML 2023:
<https://doi.org/10.1109/ICICML60161.2023.10424889>

Capstone: Solar Panel Validation Device

Jan 2019 - Apr 2019

Advisor: Haskell Jac Fought

OSU

- Developed a Bluetooth serial communication module on Android Studio.
- Designed software requirements, features and behavior charts.
- Converted Arduino coding to Android.

Advanced Energy Vehicle Design-and-Build Project

Jan 2016 - May 2016

Advisor: Lauren Corrigan

OSU

- Coded Arduino through the Arduino Integrated Development Editor.
- Used SolidWorks to design Arduino components with tightly controlled size and shape.
- Recorded Arduino performance and helped improve its aerodynamic efficiency.

CO-OP & PART-TIME

NEU Machine Learning Course

Jan 2025 - May 2025

Teaching Assistant

Maine, USA

- Guided students through various machine learning models and their corresponding techniques.
- Aligned with the professor on course allocation; held office hours for student questions and graded assignments.

OSU ECE Department Laboratory

Sep 2018 - Dec 2018

Monitor & Grader

Ohio, USA

- Guided students in coding with TI MSP430 series microcontrollers.
- Reviewed weekly quizzes and lab reports.

Huahong Jitong Smart System

May 2018 - Jul 2018

Software Development Engineer

Shanghai, China

- Mastered the architecture, data flow, operating principles and coding standards of the Automatic Fare Collection (AFC) system.
- Improved the UI design and updated the traffic data for the Xi'an Metro Line 1 to Line 4.
- Contributed to the database migration from Oracle SQL to DB2.

CEFC China Energy

May 2017 - Jul 2017

Information System Engineer

Shanghai, China

- Improved the company's front-end OA system by adding a human-friendly shift schedule and integrating it with the corporate app (Android / iOS).

- Designed an Access VBA instruction set to automatically calculate departmental phone bills with filtering, tabulation and export functions.

SKILL

Languages	Python, C#, Java, C++, Golang, Shell, JavaScript, M, VBA, VHDL
ML Frameworks	PyTorch, TensorFlow, Keras, Scikit-learn, LangGraph
Models & AI Tools	Claude, Codex, DeepSeek, Qwen, Pi, Wan, Flux, Kling, Seedance
Agent Frameworks	MCP Server, Agent Runtime / daemon, harness rule orchestration, skill library
Agent Tools	Multica, Metathinking, Paseo, Sol-Pi, HiClaw, PaperClips, Hermes
Agent Engineering	Token cost governance, model routing, prompt-cache alignment, context compaction, multi-agent orchestration
Model Optimization	Knowledge Distillation, Quantization, Pruning, Inference Acceleration
Databases	Microsoft SQL, KQL, MySQL, MongoDB, SQLite
Cloud	Azure, AWS, Distributed Systems, Microservices
Dev Tools	VS, VS Code, Git, Docker, SQL Server, Kusto, TailScale, Coder Workspace
Hardware	Arduino, Multiplexers, PLDs, Clocked Sequential Circuits, VLSI

COURSEWORK

Computer	Machine Learning, Database Management Sys, Algorithms, Cloud Computing, Software Development & Design, Operating Systems, Computer Architecture & Design, Computer Organization & Embedded Systems
Electrical	Analog Systems & Circuits, Signal & Systems, Digital Logic, Microcontrollers, Semiconductor Electronic Devices, Optical Engineering, Feedback Control Systems, Advanced Digital Design, VLSI Digital Signal Processing Systems
Math	Calculus for Scientists & Engineers, Linear Algebra, Higher Mathematics, Numerical Analysis, Partial Differential Equations
Others	Engineering Economic Decisions, Engineering Ethics, Probability and Statistics

HOBBIES & INTERESTS

Outside of work, I enjoy a balanced mix of activities. On the active side, I'm passionate about pickleball and badminton, where I love the fast rallies and team dynamics. On the intellectual side, I'm an avid fan of escape rooms, murder mystery games, and board games—immersing myself in rich narratives and strategic thinking. These hobbies have taught me that great AI products, much like great game design, require deep empathy for users and meticulous attention to detail. Looking forward to connecting with colleagues over a match or a game session outside of work.